Data Structures and Algorithms -3 (25CS2103E)

Campus Career Hub

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Abstract

Campus Career Hub is a web-based platform that unifies academic learning and career development, so students no longer juggle separate tools for study material, resumes, and job search.

- Smart Notes Search Engine – keyword search across notes, PYQs, assignments and lecture material, pinpointing the exact page.
- Resume Analyzer – evaluates resumes, flags missing skills and gives an ATS compatibility score.
- Job & Internship Matching – compares student skills with job requirements to recommend suitable roles.
- Portfolio Showcase – lets students display projects and GitHub repos for recruiters to review.
- Built with efficient DSA – Tries, Hash Tables, Inverted Indexing, Graphs, Priority Queues and Pattern Matching – for fast search and smart recommendations.
- Tech stack: React, FastAPI and PostgreSQL, delivering a scalable, secure, user-friendly platform.

Introduction

- Students today rely on multiple disconnected platforms for learning resources, resume building, and job or internship search — the process is fragmented and time-consuming.
- There is no single system that connects academic performance with career readiness.
- Campus Career Hub bridges this gap with one centralized platform for:
  - Finding study material instantly through intelligent search
  - Building and validating a job-ready resume
  - Discovering relevant jobs and internships
  - Showcasing academic and project work to recruiters and faculty
- The system is powered by well-chosen data structures and algorithms, ensuring speed and scalability as data grows.

Literature Survey

- Existing job portals (LinkedIn, Naukri, Indeed) focus only on job discovery — they have no link to a student's academic notes or coursework.
- Resume screening tools (Zety, Resume Worded) score resumes but work in isolation, disconnected from a student's verified academic record.
- College LMS platforms (Moodle, Google Classroom) store notes and assignments but offer only basic keyword search with no page-level retrieval.
- Generic search engines are not tuned for structured academic content such as PYQs, notes, and assignments.

- **Research gap identified: no existing system unifies fast academic content search, resume evaluation, and job/internship matching using efficient DSA-driven techniques in one platform — the motivation for Campus Career Hub.**

Problem Statement and Objectives

Problem Statement

Students depend on fragmented platforms for notes, resumes and job search, which wastes time and weakens career preparedness.

Objectives

- Build a Smart Notes Search Engine using Tries and Inverted Indexing for fast, page-accurate retrieval.
- Develop a Resume Analyzer that detects missing skills and computes an ATS compatibility score.
- Design a Job & Internship Matching module using Graphs and Priority Queues to rank the best-fit opportunities.
- Enable a Portfolio Showcase for projects and GitHub repositories, visible to recruiters and faculty.
- Ensure the platform is scalable, secure and responsive using React, FastAPI and PostgreSQL.

Flow Chart

Student Path

Student Login / Sign-up

↓

Choose Module
(Notes Search / Resume / Jobs / Portfolio)

↓

Query Processed via DSA Engine
(Trie, Inverted Index, Graph, Priority Queue)

↓

Ranked Results & Job Matches Returned

↓

Student Applies to Part-Time Job & Tracks Application

Skill-fit match ⇢

Recruiter / Entrepreneur Path

Recruiter / Entrepreneur Login / Sign-up

↓

Post Part-Time Job
(role, skills needed, hours, pay)

↓

Job Indexed via DSA Engine
(Graph + Priority Queue Matching)

↓

Matched Student Profiles Ranked by Skill-Fit

↓

Recruiter Reviews Resume & Portfolio, Shortlists Student

Results

- Smart Notes Search returns keyword matches with exact page numbers in under a second, using Trie-based prefix lookup and inverted indexing.
- Resume Analyzer produces an ATS compatibility score along with a clear list of missing skills for each uploaded resume.
- Job & Internship Matching ranks opportunities by skill-fit score, computed via graph-based matching and priority queues.
- Portfolio Showcase pages successfully display student projects and linked GitHub repositories to recruiters and faculty.
- Initial testing shows the platform comfortably handles concurrent searches while keeping response times low.

Hardware and Software requirements

Hardware Requirements

- Processor: Intel i5 / AMD Ryzen 5 or higher
- RAM: 8 GB minimum (16 GB recommended)
- Storage: 256 GB SSD or higher
- Stable internet connection

Software Requirements

- Frontend: React.js
- Backend: FastAPI (Python)
- Database: PostgreSQL
- OS: Windows 10/11, Linux or macOS
- Tools: VS Code, Git & GitHub, Postman

Conclusion

- Campus Career Hub successfully unifies academic resource search and career development into one platform, removing the need for multiple disconnected tools.
- Efficient data structures — Tries, Hash Tables, Inverted Indexing, Graphs, and Priority Queues — power fast search, accurate resume evaluation, and smart job matching.
- The React, FastAPI and PostgreSQL stack makes the platform scalable, secure and easy to use.
- Overall, the system improves academic performance and career readiness while easing the transition from education to employment.

Future Scope

- AI-based chatbot for personalized career and study guidance.
- Mobile application for on-the-go access to notes, resumes and job alerts.
- Deeper integration with LinkedIn and other job platforms for real-time listings.
- Recruiter dashboard with analytics on candidate skill trends.
- Peer collaboration and mentorship features connecting juniors with alumni.
- Video-based mock interview practice with AI feedback.

Thank you